

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

# Cybercrime and Engineer's Ethical Role CASE STUDY

PRESENTED BY:

HUZAIFA SHAHBAZ[21-CE-009]

AQSA SHAH[21-CE-039]

SUBMITTED BY:

MAAM RUQYA ZUBAIR

# OUTLINES

- ABSTRACT
- INTRODUCTION
- OBJECTIVES
- HISTORY OF CYBERCRIME
- ENGINEER'S ETHICAL ROLE IN CYBERCRIME MITIGATION
- PEC CODE OF ETHICS
- RELEVANCE PEC CODE OF CONDUCT ARTICLE
- WANNA CRY RANSOMWARE ATTACK(2017)
- ENGINEER'S ETHICAL ROLE
- CONCLUSION

# OUTLINES

- LESSON LEARNED
- CONCLUSION

# ABSTRACT

- Digital Technology Advancement and Engineers
- Rapid technology advancements.
- Cybercrime and ethical challenges.
- Engineers' role in system security and integrity.

# INTRODUCTION

- Rise of cybercrime in Digital Age.
- Role of engineers in secure systems.
- Ethical responsibility in addressing cyber threats.

# Objectives

- Explore forms of cybercrime (data breaches, hacking, etc.).
- Identify engineers' roles in secure system development.
- Examine PEC Code of Ethics regarding public safety.
- Provide guidelines for ethical decision-making.
- Compare ethical vs. unethical engineering practices.
- Study engineers' role in cybercrime awareness.
- Recommend implementation of cyberspace ethics.

# History of Cybercrime

- Computer Viruses and Malware Origins
- Originated in 1960s to 1980s.
- Emergence of computer viruses and malware in 1980s.
- Notable milestones : online banking crime(1994), Love Bug Virus (2000), WannaCry(2017)



# Engineer's Ethical Role in Cybercrime Mitigation

- Adherence to PEC Code of Ethics.
- Ethical dilemmas in cybersecurity.
- Importance of justice, honesty, and accountability.

## PEC CODE OF ETHICS

- Engineers' priority: safety, health, and welfare of the public.
- Importance of compliance with data protection laws.
- Role in developing secure systems.

## **Relevant PEC Code of Conduct Articles**

- **Article 1: Honesty and Integrity**
- **Article 2: Competency**
- **Article 3: Public Interest**
- **Article 5: Confidentiality**
- **Article 7: Social Responsibility**

# WANNACRY RANSOMWARE ATTACK(2017)

- It occurred in May 2017.
- And it happened to be one of the most widespread cyberattacks in history.
- Targeted systems using Microsoft Windows.
- For encrypted files, it demands ransom in Bitcoin.
- Over 200,000 computers in 150 countries were affected.
- Caused massive disruption in hospitals, businesses, and public services.

# ENGINEER'S ETHICAL ROLE

- **Prevention and Mitigation**

- Make the software system safe by reducing the vulnerabilities.
- Regularly patch up and update the systems against exploitation.

- **Accountability**

- Engineers should take responsibility for using secure design and testing practices.
- Act according to ethical guidelines that will protect users from threats themselves.

# ENGINEER'S ETHICAL ROLE(CONT)

- Awareness and Advocacy
  - Organizations should be educated on the importance and value of cybersecurity.
  - Ethical usage of technology to avoid misuse should be encouraged.

# LESSON LEARNED

- **Importance of Ethical Engineering**

- Engineers make a significant contribution to the design of systems regarding public safety and data integrity.

- **Global Collaboration**

- Government, engineers, and authority bodies should collaborate to limp over future cyber challenges.

# Conclusion

- Engineering ethics guide engineers in tackling cybercrime.
- Importance of public interest and confidentiality.
- Engineers' role in promoting ethical practices and public trust.
- The WannaCry attack did show one serious lack: the call for ethical engineering practices and global cooperation for cybersecurity.





**THANK  
YOU**